

Personal Details

19/04/2026

Contact	j.l.cremer@tudelft.nl	LinkedIn: www.linkedin.com/in/jochen-cremer
Birth	28/07/1990 in Frankfurt am Main, Germany	Proficient in languages: German and English
Address	Den Haag, Netherlands	

Qualifications

2017-2020	PhD Applied Machine Learning for Energy Systems Management (distinction: top thesis out of 50) <i>Imperial College London, UK</i>	
2014-2016	MSc Chemical Engineering (Grade: 1.5) <i>RWTH Aachen University, Germany</i>	
2011-2016	BSc Electrical Engineering (Grade: 2.6, top 5% in 2012) <i>RWTH Aachen University, Germany</i>	} only top <1% of students study simultaneous two degrees
2010-2014	BSc Mechanical Engineering (Grade: 1.9, top 5%) <i>RWTH Aachen University, Germany</i>	

Experience

Since 12/2020	Associate Professor, Delft University of Technology (TU Delft), Netherlands Co-Director, Delft AI Lab for Electric Sustainable Energy - Developing a new research line on Artificial Intelligence for Energy Systems - Leading a team of PhDs and Postdoctoral researchers
Since 09/2023	Principal Scientist, Center for Energy, Austrian Institute of Technology Developing a team on computational methods for grid operation, planning, including techniques from Artificial Intelligence and Quantum Computing, responsible for a program with 6 PhDs + 1 Postdoc
Since 10/2021	Member Board of Directors, Student Energy, NGO, Canada Overseeing operations and the strategy of this charity with 20 FTEs, 1M budget, 50k registered students, and several programs for skill-building and development of youth, https://studentenergy.org/
2017-2021	Research Associate & PhD Student, Imperial College London, UK - Fully funded scholarship from RTE, the French TSO, and EPSRC, UK - Conference Chair of International Student Energy Summit 2019 - flagship event of Canadian charity Student Energy, London, UK www.StudentEnergySummit2019.com - Globally largest student-run sustainable energy conference (700 delegates from 100 countries) - Over 50 top-tier C-level energy leaders spoke at our 4-day conference - My role: led a 30-person team and 70 volunteers, project manager, fundraised budget of £500k, managed the international advisory board, oversaw marketing, program, logistics, moderation, etc.
2016	Graduate student researcher at Massachusetts Institute of Technology, USA 6 months master thesis on control systems in chemical engineering
2015-2016	Student researcher at E.ON Energy Research Centre and RWTH Aachen University, Germany
2013-2014	Exchange student and researcher at Carnegie Mellon University, USA 9 months of studies and bachelor thesis "Demand side management of air separation plants"
2016	Industrial placements (3 months full-time each), Germany, China Business development with CEO for medium-sized SME in energy industry, <i>Ene't, Hückelhoven, Germany</i>
2015	Project engineering intern "Explore Together" program, BASF, Shanghai, China
2012	Production engineering intern, ThyssenKrupp Steel Europe, Germany
2011	Essential engineering intern, INEOS, Cologne, Germany

Awards and Recognitions (based on merit)

2024	The Best Technical Brochure Award, IEEE PES Subcommittee AMPS (IEEE PES-TR104 in the list below)
2024	The Outstanding Reviewer, IEEE Transactions on Sustainable Energy, IEEE PES
2022	The Best Paper of IEEE Journal of Modern Power Systems and Clean Energy (J14 in the list below)
2021	The Eryl Cadwallader Davies Prize 2020-2021, Electrical Engineering, Imperial College The top "Outstanding PhD Thesis" out of 50 PhD theses in the academic year.
2019	The John Lever Memorial Award 2019 (£3k), Faculty of Engineering, Imperial College "contributed significantly to teaching, outreach, public engagement, enterprise or student experience."
2018	The John & Frances Jones Prize 2017-2018, Imperial College London "the postgraduate student who 'made the best all-round contribution to College life, taking into account academic achievement, social and extra-curricular activities, for the academic year in question'"

2017	The Best Paper Award , <i>6th International Conference on Smart Cities and Green ICT, Portugal</i>
2015	The Research Ambassador Award (£2k) for highly qualified master's students, <i>RWTH Aachen</i>
2012-2016	The Germany Scholarship (£11k) for academic achievements by <i>Evonik Foundation, Bilfinger Berger</i>
2013-2014	The Dean's List Award for academic achievements, <i>Carnegie Mellon University, USA</i>
2013-2014	The International Study and Education Partnerships Grant (worth £60k) to study in the USA by the <i>German Academic Exchange Service (DAAD)</i>

Journal Publications (* in review, ** in preparation)

-
- J46 Jochen L. Cremer, "Assessing Maintenance of Medium Voltage Cable Networks under Time-Varying Loading", *IEEE Transactions on Power Delivery*, 2026
 - J45 Runyao Yu, Viviana Kleine, Thomas Rudolf, Adrian Eisenmann, Philipp Gromotka, Gautham Ram, Peter Palensky, Jochen L. Cremer, "Probabilistic Hysteresis Factor Prediction Under Heterogeneous Driving Cycles: A Case Study on Porsche Electric Vehicles", *IEEE Transactions on Transportation Electrification*, 2026, <https://arxiv.org/html/2603.09103v1>
 - J44 Ali Rajaei, Jochen L. Cremer, "Security-Constrained Substation Reconfiguration Considering Busbar and Coupler Contingencies", *IEEE Transactions on Power Systems*, 2026, <https://arxiv.org/pdf/2603.04203>
 - J43 Jochen L. Cremer, "Customising Electricity Contracts at Scale with Large Language Models", *IEEE Transactions on Power Systems* 2026, <https://arxiv.org/abs/2505.19551>
 - J42 Luca Hofstadler, Catalin Gavriluta, Jochen Stiasny, Matthias Möller, Carsten Hartmann, Andrea Benigni, and Jochen L. Cremer, "A Compact QUBO Formulation for Solving Transmission Expansion Planning", *Electric Power System Research, and 2026 IEEE Power System Computation Conference*
 - J41 Runyao Yu, Ruochen Wu, Yongsheng Han, Jochen L. Cremer, "Orderbook Feature Learning and Asymmetric Generalization in Intraday Electricity Markets", *Electric Power System Research, and 2026 IEEE Power System Computation Conference*
 - J40 Ali Rajaei, Peter Palensky, Jochen Cremer, "Transferable Graph Learning for Transmission Congestion Management via Busbar", *Electric Power System Research and 2026 IEEE Power System Computation Conference*
 - J39 Basel Morsy, Jochen Stiasny, Adolfo Anta, Jochen Cremer "Configure-and-Bound: A fast heuristic for network topology reconfiguration", *International Journal of Electrical Power Systems, Vol. 174*, 2026
 - J38 Demetris Chrysostomou, Jose Rueda Torres, Jochen L. Cremer "Machine Learning-based Method to Support TSO-DSO Adaptive Coordination with Active Power Management for Instability Prevention", *International Journal of Electrical Power Systems, Volume 173*, 2025
 - J37 J. L. Cremer, "N-Security Assessment With Quantum Annealing," in *IEEE Transactions on Power Systems*, vol. 41, no. 2, pp. 1484-1497, March 2026, doi: 10.1109/TPWRS.2025.3618916.
 - J36 Demetris Chrysostomou, Jose Rueda Torres, Jochen L. Cremer, "TensorConvolutionPlus: A Python package for distribution system flexibility area estimation" *Software X*, 2025
 - J35 Basel Morsy, Matthew Deakin, Adolfo Anta, Jochen L. Cremer, "Corrective Substation Reconfiguration and Busbar Splitting in Hybrid AC/DC Grids", *International Journal of Electrical Power Systems, Vol 169*, 2025
 - J34 A. Rajaei, O. Arowolo and J. L. Cremer, "Learning-Accelerated ADMM for Stochastic Power System Scheduling With Numerous Scenarios," in *IEEE Transactions on Sustainable Energy*, vol. 16, no. 4, pp. 2701-2713, Oct. 2025, doi: 10.1109/TSTE.2025.3562640.
 - J33 Mert Karaçelebi, Jochen L. Cremer "Online Neural Dynamics Forecasting for Power System Security", *International Journal of Electrical Power & Energy Systems, Vol 167*, 2025
 - J32 Olayiwola Arowolo, Jochen Stiasny, Jochen Cremer, "Exploring the Extrapolation Performance of Machine Learning Models for Power System Time Domain Simulations", *12th Bulk Power System Dynamics and Control Symposium and Sustainable Energy, Grids and Networks Journal*, 2025
 - J31 Mert Karaçelebi, Jochen L. Cremer, "Power system frequency monitoring and emergency control with neural ordinary differential equations", *12th Bulk Power System Dynamics and Control Symposium and Sustainable Energy, Grids and Networks Journal*, 2025
 - J30 Geert Jan Meppelink, Ali Rajaei, Jochen L. Cremer, "A Hybrid Curriculum Learning and Tree Search Approach for Network Topology Control" *Electric Power System Research*, 2025
 - J29 Jochen L. Cremer, "Polynomial Line Outage Distribution Factors for Estimating Expected Congestion and Security", *IEEE Transactions on Power Systems*, 2024
 - J28 D. Chrysostomou, J. L. R. Torres and J. L. Cremer, "Tensor Convolution-Based Aggregated Flexibility Estimation in Active Distribution Systems," in *IEEE Transactions on Smart Grid*, vol. 16, no. 1, pp. 87-100, Jan. 2025, doi: 10.1109/TSG.2024.3453667
 - J27 Jingwei Dong, Haiwei Xie, Yucheng Liao, Jochen Cremer, and Peyman Mohajerin Esfahani, "Real-Time Ground Fault Detection for Inverter-Based Microgrid Systems", *IEEE Transactions on Control Systems*, 2024
 - J26 Haiwei Xie and Jochen L. Cremer, "Game-Theoretic Learning for Power System Dynamic Ancillary Service Provisions," in *IEEE Control Systems Letters*, vol. 8, pp. 1307-1312, 2024,
 - J25 Bastien Giraud, Ali Rajaei, Jochen L. Cremer "Constraint-Driven Deep Learning for N-k Security Constrained Optimal Power Flow", *Electric Power System Research and 2024 IEEE Power System Computation Conference*
 - J24 Nikolina Čović, Jochen L. Cremer, Hrvoje Pandžić, "Learning a reward function for user-preferred appliance scheduling" *Electric Power System Research and 2024 IEEE Power System Computation Conference* <https://arxiv.org/pdf/2310.07389.pdf>

- J23 Charles Renshaw-Whitman, Viktor Zobernig, Jochen L. Cremer, Laurens de Vries, "Non-stationarity in multiagent reinforcement learning in electricity market simulation", *Electric Power System Research and 2024 IEEE Power System Computation Conference*
- J22 Al-Amin Bugaje, Jochen L. Cremer, Goran Strbac, "Generating Quality Datasets for Real-Time Security Assessment: Balancing Historically Relevant and Rare Feasible Operating Condition" *International Journal of Electrical Power & Energy Systems*, 2023
- J21 B. Habib, E. Isufi, W. v. Breda, A. Jongepier and Jochen L. Cremer, "Deep Statistical Solver for Distribution System State Estimation" *IEEE Transactions on Power Systems*, 2023, doi: 10.1109/TPWRS.2023.3290358.
- J20 Dariush Wahdany, Carlo Schmitt, Jochen L. Cremer, "More than Accuracy: End-To-End Wind Power Forecasting that Optimises the Energy System", *Electric Power System Research*, 2023
- J19 Nidarshan Veera Kumar, Jochen L. Cremer, Marjan Popov, "Dynamic Incremental Learning for real-time disturbance event classification", *International Journal of Electrical Power & Energy Systems*, 2023, (108988)
- J18 Al-Amin Bugaje, Jochen L. Cremer, Goran Strbac, "Real-time Transmission Switching with Neural Networks" *IET Generation, Transmission & Distribution*, 2022
- J17 Al-Amin Bugaje, Jochen L. Cremer, Goran Strbac, "Split-based Sequential Sampling for Realtime Security Assessment", *International Journal of Electrical Power & Energy Systems*, 2022
- J16 Federica Bellizio, Jochen L. Cremer, Goran Strbac, "Transient Stable Corrective Control Using Neural Lyapunov Learning", *IEEE Transactions of Power Systems*, 2022
- J15 Antoine Marot, Benjamin Donnot, Karim Chaouache, Adrian Kelly, Qiuhua Huang, Ramij-Raja Hossain, and Jochen L. Cremer. "Learning to run a power network with trust." arXiv preprint arXiv:2110.12908. *Electric Power Systems Research*, 2022
- J14 Antoine Marot, Adrian Kelly, Matija Naglic, Vincent Barbesant, Jochen Cremer, Alexandru Stefanov and Jan Viebahn, "Perspectives on Future Power System Control Centers for Energy Transition", *IEEE Journal of Modern Power Systems and Clean Energy*, 2022
- J13 Federica Bellizio, Wangkun Zu, Dawei Qiu, Yujian Ye, Dimitrios Papadaskapoulos, Jochen L. Cremer, Fei Teng, Goran Strbac, "Transition to Digitalized Paradigms for Security Control and Decentralized Electricity Market", *IEEE Proceedings, Special Issue "The Evolution of Smart Grids"*, 2022
- J12 Federica Bellizio, Al-Amin B. Bugaje, Jochen L. Cremer, Goran Strbac, "Verifying Machine Learning Conclusions for Securing Low Inertia Systems", *Sustainable Energy, Grids and Networks*, 2022
- J11 Felix Rafael Segundo Sevilla, Yanli Liu, Emilio Barocio, Petr Korba, Manuel Andrade, Federica Bellizio, Jorrit Bos, Balarko Chaudhuri, Hector Chavez, Jochen Cremer, Robert Eriksson, Camille Hamon, Miguel Herrera, Marnick Huijsman, Michael Ingram, Danny Klaar, Venkat Krishnan, Jorge Mola, Marcos Netto, Mario Paolone, Panagiotis Papadopoulos, Miguel Ramirez, Jose Rueda, Walter Sattinger, Vladimir Terzija, Simon Tindemans, Alberto Trigueros, Yajun Wang, Junbo Zhao. "State-of-the-art of data collection, analytics, and future needs of transmission utilities worldwide to account for the continuous growth of sensing data". *International Journal of Electrical Power & Energy Systems*, 2021: 107772.
- J10 F. Bellizio, J. L. Cremer, G. Strbac, "Machine-learned security assessment for changing system topologies." *International Journal of Electrical Power & Energy Systems* 134. 2022: 107380.
- J9 F. Bellizio, J. L. Cremer, M. Sun, G. Strbac, "A causality-based feature selection approach for data-driven dynamic security assessment." *Electric Power Systems Research* 201. 2021: 107537.
- J8 A. Bugaje, J. L. Cremer, M. Sun, G. Strbac, "Selecting decision trees for power system security assessment", *Energy and AI*, 2021: 100110.
- J7 T. Zhang, M. Sun, J. L. Cremer, N. Zhang, G. Strbac, C. Kang, "A Confidence-Aware Machine-Learned Framework for Dynamic Security Assessment", *IEEE Transactions on Power Systems*, 2021
- J6 J. L. Cremer, G. Strbac, "A Machine-learning based Probabilistic Perspective on Dynamic Security Assessment" *International Journal of Electrical Power & Energy System*, 2020
- J5 J. L. Cremer, I. Konstantelos, G. Strbac, "From Optimization-based Machine Learning to Interpretable Security Rules for Operation", *IEEE Transactions on Power Systems*, 2019
- J4 M. Pau, J. L. Cremer, F. Ponci, A. Monti, "Day-Ahead Scheduling of Electric Heat Pumps for Peak Shaving in Distribution Grids". In: *Smart Cities, Green Technologies, and Intelligent Transport Systems. Communications in Computer and Information Science, Springer Journal*, 2018
- J3 J. L. Cremer, I. Konstantelos, S. H. Tindemans, G. Strbac, "Data-driven Power System Operation: Exploring the Balance between Cost and Risk", *IEEE Transactions on Power Systems*, 2018
- J2 M. Sun, J. L. Cremer, G. Strbac, "A novel data-driven scenario generation framework for transmission expansion planning with high renewable energy penetration", *Applied Energy*, 2018
- J1 Q. Zhang, J. L. Cremer, I. E. Grossmann, A. Sundaramoorthy, J. M. Pinto, "Risk-based integrated production scheduling and electricity procurement for continuous power-intensive processes", in *Computers & Chemical Engineering*, 2015

Conference Publications (* in review, ** in preparation)

- C21 Jochen Stiasny, Jochen Cremer, "Physics-Informed Non-Dimensionalisation for Neural Solvers", *PowerUp, Boulder USA*, 2026
- C20 Ali Rajaei, Olayiwola Arowolo, Jochen L. Cremer "Spatio-Temporal Graph Neural Networks for Multi-Period Optimal Power Flow", *IEEE ISGT Europe 2025, Malta*
- C19 Jesse J. Heckman, Florian de Koning, Luc Nies, Jochen L. Cremer, "FlexFinder: Facilitating the Contracting Process for Medium Voltage Grid Customers Using Large Language Models", *CIREN 2025, Geneva, Switzerland*

- C18 Luca Hofstadler, Catalin Gavrilita, Jochen Stiasny, and Jochen Cremer, "Exploratory Analysis on the Impact of Grid Tariffs in Transmission Expansion Planning", *Kiel, Germany, IEEE PES Powertech 2025*
- C17 Thomas Lautenbacher, Ali Rajaei, Davide Barbieri, Jan Viebahn, Jochen L. Cremer, „Multi-Objective Reinforcement Learning for Power Grid Topology Control", *Kiel, Germany, IEEE PES Powertech 2025*
- C16 Catarina Santos Neves, Nikolina Čović, Jochen Cremer, "Learning to control a battery through reinforcement: balancing lifetime and profit", *Kiel, Germany, IEEE PES Powertech 2025*
- C15 Oscar Lopez Romero, Jan Poland, Jochen L. Cremer. "Deep Diffusion on Sky Images for Probabilistic Ultra-Short-Term PV Forecasting", *Kiel, Germany, IEEE PES Powertech 2025*
- C14 Viktor Zobernig, Sarah Fanta, Stefan Strömer, Regina Hemm, Jochen Stiasny, Jochen L. Cremer, Laurens J. de Vries, "Exploring Market Designs for Enhanced Flexibility Procurement with Deep Reinforcement Learning", *The International Conference on European Energy Markets, Lisboa, Portugal 2025*
- C13 Runyao Yu, Jiaqi Wang, Yongsheng Han, Chi Zhang, Teddy Szemberg O'Connor, Jochen L. Cremer, "Few-Shot Transfer Learning for Battery Cycle Life", *IEEE ISGT Europe 2024, Dubrovnik, Croatia*
- C12 J. L. Cremer, A. Kelly, R. Jorge Bessa, M. Subasic, P. Papadopoulos, S. Young, A. Sagar, A. Marot, "A Pioneering Roadmap for ML-Driven Algorithmic Advancements in Electrical Networks", *IEEE ISGT Europe 2024, Dubrovnik, Croatia*
- C11 F. R. Segundo Sevilla, Y. Liu, E. Barocio, P. Korba, A. Zamora, D. Dotta, F. Bellizio, H. Chavez, H. Jóhannsson, J. Cepeda, J. Cremer, J. Zhao, M. Arrieta, M. Paolone, M. Ramirez, P. Papadopoulos, and Y. Susuki, "Spatio-temporal Data-Driven and Machine Learning based Applications for Transmission Systems" *IEEE PES General Meeting 2024*
- C10 A. Neagu, D. Chakravorty, J. L. Cremer, "Automatic Detection of Subsynchronous Oscillations" *CIGRE Annual meeting 2024*
- C9 Demetris Chrysostomou, Jose Rueda Torres, Jochen L. Cremer "Operational Flexibility for Active Distribution Networks with low observability", *IEEE Powertech 2023, Belgrade, Serbia*
- C8 Rushil Vohra, Ali Rajaei, Jochen L. Cremer "End-to-End Learning with Multiple Modalities for System-Optimised Renewables Nowcasting", *IEEE Powertech 2023, Belgrade, Serbia*
- C7 Jasper van Tilburg, Luciano Cavalcante Siebert, Jochen L. Cremer, "MARL-iDR: Multi-Agent Reinforcement Learning for Incentive-based Residential Demand Response" *IEEE Powertech 2023, Belgrade, Serbia*
- C6 Haiwei Xie, Federica Bellizio, Jochen L. Cremer, Goran Strbac, "Regularised Learning with Physical Equations for Uncertain Power System Dynamics", *IEEE Powertech 2023, Belgrade, Serbia*
- C5 J. L. Cremer, I. Konstantelos, G. Strbac, „Optimized Operation Rules for Imbalanced Classes", *IEEE PES General Meeting of Power Systems, Atlanta, USA, 08/2019*
- C4 J. L. Cremer, I. Konstantelos, S. H. Tindemans, G. Strbac, "Sample-Derived Disjunctive Rules for Secure Power System Operation", in *IEEE International Conference on Probabilistic Methods Applied to Power Systems, Boise, USA, 06/2018*
- C3 M. Pau, J. L. Cremer, F. Ponci, A. Monti, "Impact of Customers Flexibility in Heat Pumps Scheduling for Demand Side Management", in *IEEE International Conference on Environment and Electrical Engineering, Italy, 06/2017*
- C2 J. L. Cremer, M. Pau, F. Ponci, A. Monti, "Optimal scheduling of heat pumps for power peak shaving and customers thermal comfort", in *International Conference on Smart Cities and Green ICT Systems, Portugal, 04/2017*
- C1 Q. Zhang, J. L. Cremer, I. E. Grossmann, A. Sundaramoorthy, J. M. Pinto. "Electric Energy Procurement for Large Industrial Consumers Under Uncertainty in Electricity Price and Product Demand", *AICHE, Annual Meeting, USA, 11/2014*

Book chapter

- Ch4 Christoph Brosinsky, Mert Karaçelebi, Jochen Cremer, "Purpose-driven Surrogate Models Using Machine Learning and Digital Twins: Monitoring and control for dynamic security." in *Monitoring & control of electrical power systems using machine learning techniques*, Elsevier, 2023

Technical reports

- 04/2026 IEEE PES-TR140, "Data-Driven Methods for Secure Low-Inertia Grids", **lead author**, by Working Group on Big Data & Analytics for Transmission Systems DOI 10.17023/d6n1-3q04
- 11/2024 CIGRE TB 946, "The impact of the growing use of machine learning/artificial intelligence in the operation and control of power networks from an operational perspective" by CIGRE WG C2.42
- 11/2022 IEEE PES-TR104, "Application of spatio-temporal data-driven and machine learning algorithms for security assessment taskforce", by IEEE AMPS taskforce on Application of Big Data Analytics on Transmission Systems for Dynamic Security Assessment, **prize for best technical brochure at IEEE PES**
- 07/2022 IEEE PES-TR100, "Present situation on data acquisition, handling, and analytics of operators of the transmission system in different countries and their future needs to cope with the continuous growth of data", by IEEE AMPS taskforce on Application of Big Data Analytics on Transmission Systems for Dynamic Security Assessment
- 04/2021 CIGRE TB 833 - Operating strategies and preparedness for system operational resilience by CIGRE WG C2.25
- 04/2021 Jochen Cremer, Federica Bellizio, Ioannis Konstantelos, Goran Strbac, Pauline Gambier-Morel, Patrick Panciatici, "Data-Driven Security Rules for Realtime System Prediction and Control" Energy Futures Lab, Imperial College London, <https://www.imperial.ac.uk/energy-futures-lab/reports/data-driven-security-rules-for-real-time-system-prediction-and-control/>

Funding

- 01/2026 Foundation Models for Electrical Grid Operations (FOME0), RVO Energie en Klimaat Onderzoek en Ontwikkeling (CO-I TU Delft 700k Euro)
- 01/2024 AI Testing and Experimentation Facility for the energy sector – bringing technology to the market AI-EFFECT (HORIZON-CL5-2024-D3-01-11) (CO-I 600k Euro)

- 09/2023 Public-private partnership allowance for R&D with Stedin (PI 500k Euro)
- 09/2022 NWO LTP Robust Programme, Lab Manager for Alliander ICAI AI for Energy Grids Lab (CO-I, 2.5M Euro)
- 04/2022 AIT research programme for 6 PhDs (PI, 300k Euro +1M Euro indirect funds at AIT)
- 10/2022 NWO Veni Talent Program, personal grant for top 5-10% (PI, 280k Euro)
- 01/2021 Delft AI Lab programme, Delft AI Energy Lab (PI for 1M Euro)

Invited keynotes and panel talks at scientific events

- 09/2026 **Industrial training**, ENTSO-e “Power System Reliability with Artificial Intelligence”
- 08/2026 **Panel speaker** talk “Bridging Language Models and SCOPF Tools: Cyber-Secure Automation of Outage Planning” in “New Trends in Cyber-Physical Security Analytics for the Smart Grids”, IEEE PES General Meeting, *Montreal, Canada*
- 08/2026 **Panel speaker** talk “Learning the Grid: From Foundation Models to Market-Aware Forecasting” in “Artificial Intelligence for Power System Planning and Operation”, IEEE PES General Meeting, *Montreal, Canada*
- 06/2026 **Panel speaker** “Integration of data-driven solutions to low inertia power systems” at panel “Application of Big Data & Analytics to maintain the secure operation of transmission systems — Advances by the IEEE WG”, IEEE SGSMA, *Santiago, Chile*
- 06/2026 **Invited Talk**, 5th DigiSect, Design and Operation of Digitalized Sector-Coupled Energy Systems, *TU Hamburg, Germany*
- 04/2026 **cresROADS** industrial training (15 attendees) at CRESYM General Assembly, *Barcelona, Spain*
- 03/2026 **Invited speaker**, “What Does It Take to Generalize in Electricity Price Forecasting?”, GridFM workshop, *Harvard, Boston, USA*
- 02/2026 **4th Champéry Power Conference** “All AI models are wrong, but some are useful for power systems”, *Champéry, Switzerland*
- 09/2025 **Keynote** DACH Energy Informatics Conference, *RWTH Aachen, Germany*
- 09/2025 **Speaker** at Global Operation System Summit, at 50Hertz, *Berlin, Germany*
- 09/2025 **Speaker** at 04th Grid Foundation Model workshop, topic: “GridFM for transients”, *Aachen, Germany*
- 06/2025 **Tutorial** at ACM E-Energy 2025, *Rotterdam, Netherlands*
- 05/2025 **Speaker** at DTU Summer School, *DTU, Copenhagen, Denmark*
- 04/2025 **cresROADS** presentation at CRESYM General Assembly, *Brussels, Belgium*
- 01/2025 **1-day course** on “AI in Power System Reliability Monitoring” at Smart and Green Energy Systems and Business Models hosted by *Tallinn University, Estonia*, <https://www.sustainalab.nl/event/grid-congestion-reshaping-future-energy-systems/>
- 12/2024 **Speaker** “Grid congestion: Reshaping Future Energy Systems”, *Sustainalab Universiteit van Amsterdam, The Netherlands*
- 11/2024 **Tutorial of CIGRE C2.42**, AI-day for grid operations, *Paris, France*
- 11/2024 **Speaker** at Energietage, *Vienna, Austria*
- 10/2024 **Tutorial**, IEEE PES & Power Africa Conference, *Johannesburg, South Africa*
- 10/2024 **Panel speaker** “ML/AI applications in power systems: Drivers and barriers” *IEEE ISGT Europe, Dubrovnik, Croatia*
- 09/2024 **Speaker** “AI for NetZero” webinar series, <https://www.youtube.com/watch?v=xpwHfdZfgk>
- 09/2024 **Speaker** at summer school „Inverter-Based Power Systems”, *Imperial College London, UK*
- 07/2024 **Panellist**, Dynamic Modelling of Multi-Energy Systems, *IEEE PES General Meeting, Seattle, USA*
- 07/2024 **Panellist** Convergence of AI- and Physics-based Approaches in Power System Analysis, Optimization, and Control, *IEEE PES General Meeting, Seattle, USA*
- 06/2024 **Invited speaker** “Reinforcement Learning for Energy Community Management” in special session RL for energy networks” at the workshop on Reinforcement Learning for Stochastic Networks (RL4SN), *Toulouse, France*
- 05/2024 **Invited speaker** Webinar at IEEE PES Chapter Spain, Women in Power, *Universitat Politècnica de Catalunya, Barcelona, Spain*
- 05/2024 **Invited speaker in technical session** “Advancing Power System Reliability with System-based Deep Learning”, Electrimacs conference, *Castellon, Spain*
- 05/2024 **Invited speaker**, Symposium “Rise of AI | Opportunity or threat?”, *Dutchpower, Amsterdam, Netherlands*
- 03/2024 **EPSRC Supergen Energy Networks Hub Risk and Resilience Day** “Constraint-Driven Deep Learning for N-k Security Constrained Optimal Power Flow”, *Newcastle University, UK*
- 02/2024 **3rd Champéry Power Conference** “Weakly-supervised Graph Neural Networks for Distribution System State Estimation”, *Champéry, Switzerland*
- 11/2023 **Keynote speaker**, 33rd Workshop on Computational Intelligence, *HTW Berlin, Germany*
- 11/2023 **Keynote speaker** “AI-based Monitoring for Sustainable Energy Systems”, Stichting Pipeliner - Symposium '23, *Gemeente Rotterdam, Netherlands*
- 08/2023 **Speaker** “Weakly supervised learning for power grid state estimation,” in session “Physics-based and data-driven modelling for digital twin.” 10th international conference on industrial and applied mathematics, *Tokyo, Japan*
- 07/2023 **Panellist** “AI to predict power blackouts” in panel session “Application of Big Data and AI/ML in monitoring, operations, planning and protection”, *IEEE PES General Meeting, Orlando, USA*
- 07/2023 **Panellist** “Graph Neural Networks for Distribution State Estimation” in panel session “Machine Learning and Modern Heuristic Optimization for Planning and Operation of Active Distribution Networks”, *IEEE PES General Meeting, Orlando, USA*

- 06/2023 **Panellist**, Panel Young Professionals Panel Session on Future Power System Workforce, *IEEE PES PowerTech, Belgrade, Serbia*
- 06/2023 **Panellist**, Panel Contemporary and emergent methods for planning and analysis of distribution networks, title “Graph Neural Networks for Distribution State Estimation”, *IEEE PES PowerTech, Belgrade, Serbia*
- 03/2023 **Round table discussion** “AI and ML for smart grids”, Smart Grid Tech Week, invited by Utilidata, Smart Grid Forums, *Amsterdam, Netherlands*
- 02/2023 **Invited speaker** “AI for Energy – Exploring activities and opportunities in the Netherlands” at a DOE-organized workshop on AI for Energy Innovation held in conjunction with 37th AAAI Conference on Artificial Intelligence, Washington, DC, US
- 10/2022 **Plenary panellist in supersession on Stability, Dynamics, and Control of Emerging Renewable-Rich Power Systems**, *IEEE ISGT Europe, Novi Sad, Serbia*
- 10/2022 **Plenary panellist in supersession on AI for active power networks**, *IEEE ISGT Europe, Novi Sad Serbia*
- 09/2022 **Keynote address** “AI for Distributed Energy Systems”, *Intelligent Systems Conference IntelliSys 2022, Amsterdam, Netherlands* <https://www.youtube.com/watch?v=80x3ebPWlA> **100 attendees**
- 07/2022 **Plenary panellist**, “Machine Learning for Security Assessments in Low Inertia Grids”, *supersession on AI applications in Power Grid, IEEE PES General Meeting, Denver, USA*
- 07/2022 **Panellist** “Multi-Agent Reinforcement Learning for Incentivized Flexibility from Residents”, in session *Advanced Applications of Modern Optimization and Artificial Intelligence Methods on Active Distribution Networks. IEEE PES General Meeting, Denver, USA*
- 07/2022 **Panellist** “Learning to run a power network with trust”, in session *Enhancing power system operation through online analytics, IEEE PES General Meeting, Denver, USA*
- 05/2022 **Panellist** “On Dynamics and Artificial Intelligence for Power Systems”, in session *Synchrophasor and Monitoring data handling: A perspective of ongoing TSOs approach, IEEE International Conference on Smart Grid Synchronized Measurements and Analytics, Split, Croatia*
- 05/2022 **Panellist** “Artificial Intelligence Methods for DSA” in session *PMU- and AI-based analysis for a resilient operation of future power systems, IEEE International Conference on Smart Grid Synchronized Measurements and Analytics, Split, Croatia*
- 10/2021 **Lightning talk**, World Summit AI, *Amsterdam, NL* <https://youtu.be/rkpWCiYNJcs> **50 attendees**
- 09/2021 **Keynote address** “AI: Real-Time Operations or Full Automation?”, *ISA Knowledge Days, Virtually, Columbia* **300 attendees**
- 08/2021 **Panellist** “Operational Resilience”, *CIGRE Annual Meeting, Online, Paris*
- 12/2019 **Presentation, IEEE Big Data & Analytics Webinar** <https://www.youtube.com/watch?v=X8VBSOE5is&t=1>
- 11/2019 **Panellist**, *IEEE Sustainable Power and Energy Conference, Beijing, China*, **100 attendees**
- 10/2019 **Keynote address** “A Student Perspective on Climate Change”, *AAPG Energy Transition Forum, Edinburgh, UK*, **300 attendees** <https://www.youtube.com/watch?v=HGAY34cgdAo&t=1s>
- 08/2019 **Panellist**, “Data Analytic Tools for Security Assessment of Bulk Power Systems”, *IEEE PES General Meeting, Atlanta USA*

Invited seminar talks

- 05/2025 Guest Lecture, *University of Liege, Liege, Belgium*
- 11/2024 Graph & Data Seminar, *Delft University of Technology, The Netherlands*
- 10/2024 Swiss Federal Laboratories for Materials Science and Technology, *Urban Energy System, Online, Zurich, Switzerland*
- 04/2024 Zagreb University of Technology “Power System State Estimation with Deep Learning”, *Zagreb, Croatia*
- 04/2024 Hitachi Energy Corporate Research, *Zurich, Switzerland*
- 04/2024 EU Joint Research Centre, *Ispra, Italy*
- 04/2024 ie3 Seminar, *Dortmund University of Technology, Dortmund, Germany*
- 08/2023 Invited to visit the University of Kyoto, *Kyoto, Japan*
- 11/2022 The Polytechnic University of Madrid, *Madrid, Spain*
- 07/2022 Hitachi Energy Corporate Research, *Mannheim, Germany*
- 03/2022 Lunch Lecture, *TU Delft Energy Club, Delft, Netherlands*
- 03/2022 The e-refinery annual workshop, *TU Delft, Netherlands*
- 12/2021 UK Digital Catapult: Accelerating UK-NL AI Collaboration towards Net Zero, *London, UK*
- 10/2021 Eindhoven University, *Cyber-Physical Systems Center, Eindhoven, Netherlands*
- 09/2021 RTE, French TSO, R&D Department, *Paris, France*
- 05/2021 Groningen University, “Smart Grids Symposium”, *Student Energy Chapter, Groningen, Netherlands*
- 05/2021 AI for Energy and Sustainability Think Tank, *TU Delft, Delft, Netherlands*
- 04/2021 „AI for the energy system“, *Karlsruhe Institute of Technology, Institute of Applied Informatics, Karlsruhe, Germany*
- 02/2021 “Supervised Learning to Prevent Blackouts in Power Systems”, *ETH Zurich, Chair of Intelligent Maintenance Systems* <https://www.youtube.com/watch?v=mB5Adxw1idc>
- 11/2020 AI Initiative Talk, *TU Delft, Delft, Netherlands*
- 12/2019 IEEE Big Data & Analytics Webinar <https://www.youtube.com/watch?v=X8VBSOE5is&t=1>
- 10/2019 Energy Futures Lab Seminar series, *Imperial College London, London, UK*

Other publicity

05/2024	Article “KI In Energietechnik”, Österreichischer Verband für Elektrotechnik e+i Magazin
11/2023	Blog post “Roadmap of Innovating with AI for Distribution Grids”, https://www.linkedin.com/pulse/roadmap-innovating-ai-distribution-grids-jochen-cremer-zugme
10/2023	Article, “Nvidia: AI solves problems of overcrowded power grid”, Telegraaf NL https://www.telegraaf.nl/financieel/815514148/nvidia-ai-lost-problemen-van-overvol-stroomnet-op
10/2022	Blog post “Discussing the opportunities and challenges for AI-based algorithms in distribution grids” https://www.linkedin.com/pulse/discussing-opportunities-challenges-ai-based-grids-jochen-cremer/
04/2022	Article “In the future, power disruptions may follow in quick succession”, https://www.delta.tudelft.nl/article/future-power-disruptions-may-follow-quick-succession
11/2021	NLAIC video contribution to “AI for Energy Transition” https://www.youtube.com/watch?v=bZaFpHjH1VQ
10/2019	Blog post “Machine-Learning, distributed data and the grid”, Energy Futures Lab, London https://energyfutureslab.blog/2019/10/17/machine-learning-distributed-data-and-the-grid/

Leading and organising technical discussions

06/2026	Panel chair, IEEE PES SGSM, Santiago, Chile
10/2025	Panel chair “Beyond the Blackout: Building Resilient Grids for an Interconnected Europe” at ISGT Europe, Malta
09/2025	Moderator session “AI-ready grid data” at 4 th Grid Foundational Model workshop, RWTH Aachen
05/2025	Organiser “Delft Symposium Power System Resilience and Security with AI”, Delft
09/2024	Co-Organiser 14. (Hybrid) Symposium Communications for Energy System (COMFOREN), Vienna
05/2024	Organiser workshop for industry, “AI use cases for distribution networks”, Alliander
03/2024	Moderator Symposium “Wide area monitoring, and data-driven aspects for the future power system” Inauguration of Prof. Marjan Popov, Delft
10/2023	Moderator panel “Roadmap of Innovating with AI for Distribution Grids”, IEEE PES ISGT Europe 2023, Grenoble France
09/2023	Chair and organiser 7 th International DynPower Workshop, TU Delft
06/2023	Co-Chair technical session “Monitoring and control”, IEEE PES PowerTech Belgrade
03/2023	Organiser workshop “GNNs and RL for Energy System Planning and Operation”, Austrian Institute of Technology, Vienna, Austria
10/2022	Panel chair “Data analytics, machine learning, and artificial intelligence in smart grids”, IEEE European Conference on Innovative Smart Grid Technologies, Novi Sad, Serbia
08/2022	Panel chair “Advanced Applications of Modern Optimization and Artificial Intelligence Methods on Active Distribution Networks”, IEEE PES General Meeting, Denver, USA
08/2022	Panel chair “Data collection and future needs to account for the continuous growth of sensing data in control rooms”, IEEE PES General Meeting, Denver, USA
2021-2024	Organizer, 50 AI for E&S Think Tank Seminar Series, (bi-)weekly TU Delft, Rotterdam, Leiden University
10/2021	Panel moderator “Action-oriented innovations from the private sector” World Summit AI, <i>Amsterdam</i>
12/2019	Workshop, “Entrepreneurship, Climate Change, Impact”, <i>University of Leeds, UK</i> ,
07/2019	Chair and moderator of the 3-day conference, International Student Energy Summit 2019 , London, UK 700 attendees

Education

2026	Training to industry, ENTSO-e
2026	Training to industry CresRoad, CRESYM
Since 2025	Lecturer in 4 ECTS BSc course EEX01 “Introduction to Machine Learning”, BSc Electrical Engineering, (~80 students)
Since 2024	Profile coordinator “Digital Energy Technology”, MSc Sustainable Energy Technologies
Since 2024	Responsible lecturer, MSc course SET3125 “Machine Learning Workflows for Energy Systems” (~20 students)
2024	Professional training, 3-day workshop “AI for Electrical Systems” for industry Alliander & Stedin
2023	Organiser, “Learning to Run a Power Network Challenge Delft” https://codalab.lisn.upsaclay.fr/competitions/12420 , National-wide PhD, Postdoc, and early-career training
2023	Organiser of 3-day workshop “AI for Intelligent Energy Systems”, TU Delft
2023-2024	Guest lecture in EE3065TU, Reliability of Sustainable Power Systems
Since 2022	Lecture “AI for Intelligent Electricity Systems” in SET3065 TU Delft
Since 2022	Coordinator and lecturer of EE 4C12 “Machine Learning for Electrical Engineering”, EE MSc course, TU Delft (~100 students)
2022	Lecture “Machine Learning for Energy Systems” in professional 1-year course X!Lead, TU Delft
2021	Coordinator, developer, lecturer of EE 4720 “Machine Learning for Power System Applications”, EE MSc course, TU Delft
2021	Tutor for Electronics Lab EPO1 “Booming Bass”, TU Delft
2021	Developer and lecturer of a module for MOOC “Intelligent and Integrated Energy Systems”, PowerWeb Institute, TU Delft
2018	Electronics Lab, Graduate Student Assistant, Imperial College London
2017	Mathematics 1-2, Graduate Student Assistant, Imperial College London

Professional certifications

11/2022	University Teaching Qualification: supervision, course development, teaching and assessment, TU Delft
2021	Personal Development Program, TU Delft

Working groups

Since 2024	Verein Deutscher Elektrotechnik (VDE), Energietechnische Gesellschaft, Gremien „KI in der Netzleittechnik“
Since 2022	IEEE PES WG, “Application of Big Data Analytics on Transmission System Dynamic Security Assessment”, Pillar lead of a sub-group “Data-Driven methods for Low-Inertia Systems”, coordinating a technical report
Since 2023	Linux Energy Foundation, working group on AI for Energy
Since 2022	CIGRE WG 2.42, “The impact of the growing use of ML/AI in operation and control of Power Networks from an Operational perspective”, NL representative
Since 2021	NL AI Coalition WG Energy and Sustainability
2019-2022	IEEE PES taskforce, “Application of Big Data Analytics on Transmission System Dynamic Security Assessment”, Active founding member, UK representative
2017-2021	CIGRE WG 2.25 “Operational Resilience”, UK representative and task-leader
2019	Climate Entrepreneurs Club, I founded this new club in London to ideate solutions to the local effects of climate change. The club is still active www.climateentrepreneurs.uk and hosts skill development and brainstorming sessions.
2018	IEEE student branch, I built a team to organize seminars and industrial visits at Imperial College London

Editorial and reviewer service

Since 2025	Associate Editor, <i>IEEE Transactions on Power Systems</i>
Since 2025	Associate Editor, <i>IEEE Power Engineering Letters</i>
2026	Technical program committee, 24th Power System Computing Conference, Cyprus
2025	Technical program committee, IEEE ISGT Europe, Malta
2025	Technical program committee, 13th DACH+ Energy Informatics Conference, Aachen, Germany
2025	Technical program committee, 16th ACM International Conference on Future and Sustainable Energy Systems (ACM e-Energy), Rotterdam
2025	Guest editor for a special issue on “VSI:Stability Analysis of transmission system”, <i>International Journal of Electrical Power Engineering</i>
2024	Technical program committee, 12th DACH+ Energy Informatics Conference, Lugano, Switzerland
2024	Technical program committee, 15th ACM International Conference on Future and Sustainable Energy Systems (ACM e-Energy), Singapore
2024	Guest editor for a special issue on “Big Data Analytics: Moving towards applications to maintain the secure operation of carbon-neutral power grids,” <i>IEEE Open Access Journal of Power and Energy</i> , IF 2.8
2023	Guest editor for a special issue on “explainable AI for smart grid” for <i>IEEE Journal of Emerging and Selected Topics in Industrial Electronics</i> , IF 4.5
2022	Technical program committee, IEEE ISGT Europe, Novi Sad, Serbia
2021	Guest editor for a special issue on “Applications of machine learning and statistical methods for dynamic security assessment on transmission systems”, <i>Sustainable Energy, Grids and Networks</i> , IF 2.5
Since 2018	Reviewer of around 100+ IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Sustainable Energy, IEEE Transactions on Energy Markets, Policy, and Regulation, International Journal of Electrical Power & Energy Systems, Electrical Power Systems Research

Funding proposal reviewer

2025	Jury Member, NWO Open Technology Program (OTP)
2024	Deutsche Forschungs Gemeinschaft (DFG)

Peer representation & community support

Since 2023	Faculty advisor to IEEE PES Student Branch Chapter, Delft University of Technology
2018-2020	Elected department PhD representative of 250 PhD and 100 masters. Led a group of 6 reps, initiated the exchange and monthly seminars, and improved the academic environment with the management of the department
2019	Established & chaired the IEEE PES Student Branch at Imperial College London

Research projects

Since 2026	FOMEIO Foundation Models for Electrical Grid Operations, Alliander, TU Delft, RVO Energie en Klimaat Onderzoek en Ontwikkeling
Since 2024	AI-EFFECT, HORIZON-CL5-2024-D3-01-11, AI Testing and Experimentation Facility for the energy sector – bringing technology to the market
Since 2023	Graph-based AI Monitoring Tools for Complex-Energy Systems, public-private partnership project, Topsector ICT
Since 2022	Austrian Institute of Technology, Principal Scientist Program, Principal Investigator
Since 2022	NWO LTP Robust Project “Trustworthy Artificial Intelligence” Co-Investigator TU Delft

Since 2022	NWO Veni Project “physics-informed AI to predict power blackouts”, Principal Investigator
2021-2025	MEGAMIND: Measuring, Gathering, Mining and Integrating Data for Self-management in the Edge of the Electricity System, NWO Perspective Program, a supervisor from TU Delft
2017-2020	Industrial R&D project, RTE, French transmission system operator
2016	EU H2020 project FLEXMETER, work package “Energy management of heat pumps for the benefit of distribution system operators”, at E.ON Energy Research Centre, Aachen
2015	Industrial project with BASF, “Design of micro chemical reactors considering energy and mass balances” at Process Systems Engineering Group

Postdoctoral researchers

Since 2026	Kutay Bölüt, “Foundation Models for Time Series Data in Power Systems”
Since 2026	Haiwei Xie, “Theory of Scaling Data Synthetisation”
Since 2025	Maosheng Yang, “Data Synthesis for Machine Learning in Power Systems”
Since 2024	Jochen Stiasny, “Learning Surrogate Models for Physical Systems”

PhD candidates as the main supervisor

Since 2025	Selin Ezgi Özcan, “Machine Learning for Distribution System State Estimation”, TU Delft, Stedin
Since 2025	Daniel Wieser, “Virtual Power Plants for Trading”, TU Delft, Austrian Institute of Technology
Since 2025	Paul Bannmüller, “Reinforcement Learning in Topology Reconfigurations”, TU Delft
Since 2024	Betül Mamudi, “Machine Learning for State Estimation”, TU Delft, Alliander
Since 2024	Luca Hofstadler, “Quantum Computing for Transmission Systems”, TU Delft, Austrian Institute of Technology
Since 2023	Runyao Yu, “Machine Learning for Outliers in Grid Operations and Intraday Energy Markets, TU Delft, Austrian Institute of Technology
Since 2023	Olayiwola Arowolo, “Graph Neural Networks for Power Systems”, TU Delft
Since 2022	Basel Morsy, “Reinforcement Learning for Topological Reconfigurations”, TU Delft, Austrian Institute of Technology
Since 2022	Viktor Zobernig, “Reinforcement Learning for Trading Strategies in Redispatch Markets”, TU Delft, Austrian Institute of Technology
2021-2026	Ali Rajaei, “System-theoretic Machine Learning for Congestion Management,” TU Delft, TU Delft AI Initiative
2021-2026	Mert Karagelebi, “Neural Networks for Dynamic Security Assessment,” TU Delft, TU Delft AI Initiative
2022-2026	Haiwei Xie, “Control of Dynamic Virtual Power Plants”, TU Delft
2021-2025	Demetris Chrysostomou, “Learning-Based Distributed Resource Flexibility Estimation for Enhanced TSO-DSO Coordination”, TU Delft, MEGAMIND program
2019-2023	Al-Amin Bugaje, “Data Augmentation for Power System Security Assessment”, Imperial College London, now at Hitachi Energy, Canada
2018-2022	Federica Bellizio, “Topological Changes in Data-Driven Dynamic Security Assessment for Power System Control”, Imperial College London, now at Swiss Federal Laboratories for Materials Science and Technology

PhDs evaluated as external reviewer

2026	Francesca Rossi, Polytechnic University of Catalonia, Spain
2026	"Data-Driven and Generative Methodologies for Enhanced Grid-Edge Visibility in Distribution Grids", Marc Jené, Polytechnic University of Catalonia, Spain
2025	“Providing Flexibility Services to the Power System Using Demand Response and Hydrogen Technologies”, Nikolina Čović, Zagreb University, Croatia
2025	“Predictive Maintenance of Medium Voltage Cable Networks using Machine Learning”, Konrad Sundsgaard, DTU Technical University of Denmark, Denmark
2024	“Segmentation of the electricity network for smart grid management”, Noureddine Henka, Université Paris-Saclay, France
2024	"Essential Tools and Strategies for the Participation of Battery Energy Storage Systems in Ancillary Services." Javier Cardo, Universitat Jaume I de Castelló, Spain
2023	“Advanced methods for enhancing interpretability of AI tools with application to the energy sector”, Konstantinos Parginos, MINES Paris, France

MSc student supervisions

2026	“Probabilistic nowcasting of residential PV fleets in the Netherlands”, Mark van der Gun, Pythia Energy and TU Delft
2026	“Deep Learning for Thermal Modelling of Transformers”, Pauline Hengst, Allinder, and TU Delft
2025	“Tap changer level estimation with Graph Neural Networks”, Soham Prajapati, Stedin and TU Delft, supervised by Tamas Keviczky
2025	“Information theory for dynamic virtual power plants” Wim Verschuren, Haiwei Xie, Mathijs de Weerd, TU Delft
2025	“Interpretable Intraday Price Prediction Integrating Kolmogorov-Arnold Networks and Large Language Models” Ruochen Wu, Runyao Yu, AIT, TU Delft
2025	“Spatio-Temporal Graph Neural Networks for Power Grid State Estimation Under Topological Changes”, Yusuwan Chang, TU Delft
2025	“ML for Dynamics in Power Systems”, Ananth Rajagopal, TU Delft

- 2024 “Algorithms for Distribution Grid Expansion”, Sven van der Voort, TU Delft, supervised in collaboration with the Algorithmic group, Neil Yorke-Smith
- 2024 “Reinforcement Learning for Congestion Management”, Thomas Lautenbacher, TU Delft and TU Berlin
- 2024 “Incentive-Based Community Management with Network Constraints”, Michal Okon, TU Delft
- 2024 “Stable Diffusion for Energy-system Optimized Solar Nowcasting”, Oscar Lopez, TU Delft and Hitachi Energy
- 2024 “User-centric EV charging cycles to maximise battery lifetime”, Viviana Kleine, TU Delft and Porsche
- 2023 “Reinforcement Learning for Coordinating Energy communities”, Catarina Santos Neves, TU Delft in collaboration with Instituto Superior Técnico, Lisbon, Portugal
- 2023 “Market mechanisms for Frequency Service provided by Dynamic Virtual Power Plants”, Torben Zeller, TU Delft in collaboration with RWTH Aachen University
- 2023 “Quantum Computing for Power Systems” Hjalmar Lindstedt, TU Delft
- 2023 “Market Mechanism Design for Virtual Inertia”, Johnny Zheng, TU Delft
- 2023 “Reinforcement learning for transmission network topology control” Geert-Jan Meppelink, TU Delft, in collaboration with NTNU, Trondheim, Norway
- 2023 “Neural Ordinary Differential Equations for Frequency Dynamics” Nila Krishnakumar, TU Delft
- 2023 “End-to-end learning for N-k SC-OPF” Bastien Giraud, TU Delft, in collaboration with NTNU, Trondheim, Norway
- 2023 “Data-Driven Adaptive Dynamic Equivalents of Active Distribution and Transmission Networks” Alex Neagu, TU Delft
- 2022 “Multi-Modal End-to-End Learning for Real-Time Monitoring of Sustainable Energy Systems”, Rushil Vohra, TU Delft
- 2022 “Meter placement for state estimation in distribution networks”, Sattama Datta, TU Delft, thesis with Alliander, DSO Netherlands
- 2022 “Deep Statistical Solver for Distribution System State Estimation,” Benjamin Habib, supervised with Multimedia Computing Group, TU Delft, thesis with Stedin, DSO Netherlands
- 2021 “Multi-agent reinforcement learning for incentive-based residential demand response”, Jasper van Tilburg, supervised in collaboration with Interactive Intelligence Group TU Delft, now Ciphix, Netherlands, **ranked as top student-led paper at IEEE PowerTech Belgrade 2023 conference** (*prize not awarded as formal requirements not met*).
- 2021 “Detection and Classification of Faults in Small-Scale Residential PV Systems with a Synthetic PV Training Database” Dion de Mooy, supervised in collaboration with PVMD group, TU Delft, now Zonnepark Services, Netherlands,
- 2020 “Application of Machine Learning for assessing grid security performance”, Haiwei Xie, Imperial College London, **awarded best MSc thesis, now PhD student at TU Delft**
- 2020 “Application of Machine Learning for assessing grid security performance”, Ailing Guo, Imperial College London, now State Grid, China
- 2019 “Security Analysis of Electrical grids using Machine Learning”, Al-Amin Bugaje, Imperial College London, now PhD student at Imperial College London

Academic visitors

- 2026 Burak Karabulut, PhD researcher Gent University, Belgium
- 2024 Thomas Lautenbacher, MSc researcher, TU Berlin, Erasmus+ Scholarship
- 2023 Torben Zeller, MSc researcher, Erasmus+ Scholarship, graduate researcher, RWTH Aachen University
- 2022 Nikolina Čović, PhD researcher, University of Zagreb, Croatia
- 2021 Dariush Wahdany, MSc researcher, RWTH Aachen, IDEA league fellowship, now PhD at Fraunhofer Institute, Berlin